

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0308	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	25-08-26

Code of Conduct

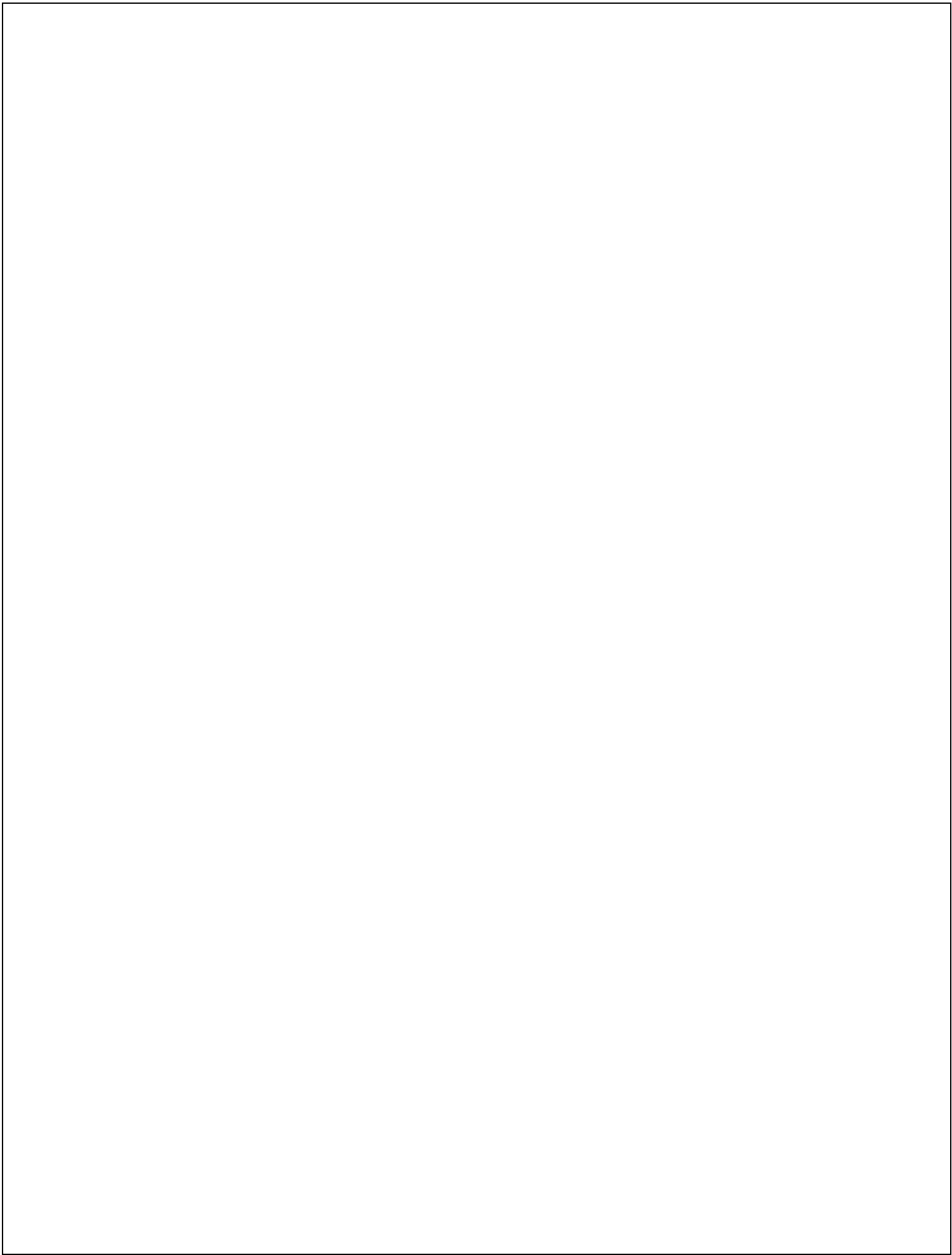
*I **M Poojitha Reddy - 192472352** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **M Poojitha Reddy**

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4



Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$
- $\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$
- $\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$
- $\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$
- $\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	4	4	3	3	18	74
2	4	3	4	4	4	19	
3	4	4	3	4	4	19	
4	4	2	4	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
T Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Frame Semantics in Banking Customer-Service Systems

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.

Answer:

Frame semantics represents the meaning of a sentence using an **action/event (frame)** and its associated **participants or semantic roles**.

a) Transfer Request

Query: “Transfer ₹20,000 from my savings account to my son's account.”

Semantic Frame:

TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)

- **Action:** Transfer
- **Agent:** Customer
- **Source:** Savings account
- **Amount:** ₹20,000
- **Destination:** Son's account

The customer initiates the transfer, money is taken from the source account, and the destination account receives it.

b) Debit Card Blocking

Query: “Block my debit card immediately.”

Semantic Frame:

BLOCK(DebitCard, Customer)

- **Action:** Block
- **Agent:** Customer
- **Object:** Debit card

The customer requests that the debit card be blocked.

c) Transaction Statement

Query: “Send my transaction statement to my email.”

Semantic Frame:

`SEND(TransactionStatement, Email, Customer)`

- **Action:** Send
- **Agent:** Customer
- **Object:** Transaction statement
- **Destination:** Email

The customer requests that the transaction statement be sent to the specified email destination.

d) Withdrawal Limit

Query: “Increase my daily withdrawal limit.”

Semantic Frame:

`MODIFY(WithdrawalLimit, Customer)`

- **Action:** Modify
- **Agent:** Customer
- **Object:** Withdrawal limit
- **Direction:** Increase

Thus, semantic frames connect the **action with its participants**, allowing the system to understand the intended banking operation.

2. Identify the query in which the semantic interpretation is incorrect and explain why.

Answer:

The incorrect interpretation is **B4**.

- **Actual User Intent:** Increase withdrawal limit
- **System Interpretation:** Decrease withdrawal limit

The system correctly identifies that the **withdrawal limit** must be modified, but it misunderstands the **direction of modification**.

The word “**increase**” means that the limit should become higher, whereas “**decrease**” means that the limit should become lower. Therefore, B4 represents an incorrect semantic interpretation.

3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.

Answer:

Incorrect semantic-role identification can result in serious banking errors. For example, if the system confuses the **source account** and **destination account**, money may be transferred from or to the wrong account.

Similarly, interpreting “**increase withdrawal limit**” as “**decrease withdrawal limit**” may unnecessarily restrict the customer's transactions. Therefore, banking systems must correctly identify roles such as **Agent, Object, Source, Destination, Amount, and Action Direction** before executing an operation.

4. Recommend a frame-based semantic analysis approach to improve reliability.

Answer:

A banking conversational system should use a **frame-based semantic parser**. First, it should identify the customer's intent, such as **TRANSFER, BLOCK, SEND, or MODIFY**, and then fill the corresponding semantic roles.

For high-risk operations, the system should validate all important fields such as account, amount, destination, and action direction. It can also ask for confirmation before execution, for example: “**Do you want to increase your daily withdrawal limit?**” This improves reliability and prevents costly banking errors.

2. First-Order Predicate Calculus for Smart Agriculture

1. Represent the field observations using First-Order Predicate Calculus.

Answer:

First-Order Predicate Calculus represents facts using **predicates, constants, variables, and logical operators**.

The field observations can be represented as follows:

Field F1

- `Dry(F1)`
- `IrrigationAvailable(F1)`
- `Rice(F1)`

Field F2

- `Wet(F2)`
- `IrrigationAvailable(F2)`
- `Wheat(F2)`

Field F3

- `Dry(F3)`
- `IrrigationUnavailable(F3)`
- `Maize(F3)`

Field F4

- `Wet(F4)`
- `IrrigationUnavailable(F4)`
- `Rice(F4)`

The given rules can be represented as:

1. $\forall x \text{ [Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)]$
2. $\forall x \text{ [NeedsWater}(x) \rightarrow \text{Irrigate}(x)]$
3. $\forall x \text{ [Wet}(x) \rightarrow \neg \text{NeedsWater}(x)]$
4. $\forall x \text{ [IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)]$
5. $\forall x \text{ [Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)]$

These rules allow the system to derive irrigation decisions automatically.

2. Using the given rules, determine which fields require irrigation.

Answer:

F1:

`Dry(F1)` and `IrrigationAvailable(F1)` are true.

Therefore:

`NeedsWater(F1)`

and:

`Irrigate(F1)`

Hence, **F1 requires irrigation and can be irrigated.**

F2:

`Wet(F2)` is true.

Therefore:

`¬NeedsWater(F2)`

Hence, **F2 does not require irrigation.**

F3:

`Dry(F3)` is true, so the field needs water from a crop-condition perspective. However:

`IrrigationUnavailable(F3)`

Therefore:

`¬Irrigate(F3)`

Hence, **F3 cannot currently be irrigated.**

F4:

Wet (F4) is true.

Therefore:

$\neg \text{NeedsWater}(\text{F4})$

Hence, **F4 does not require irrigation.**

Final Result:

Field Decision

F1 Requires irrigation

F2 Does not require irrigation

F3 Needs water but irrigation unavailable

F4 Does not require irrigation

3. Determine whether F1 and F3 should be treated differently.

Answer:

Yes, **F1 and F3 should be treated differently** even though both fields have dry soil.

For F1:

$\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1}) \rightarrow \text{NeedsWater}(\text{F1})$

Therefore:

$\text{Irrigate}(\text{F1})$

For F3:

$\text{Dry}(\text{F3}) \wedge \text{IrrigationUnavailable}(\text{F3})$

Therefore, although F3 needs water, the controller must not activate irrigation because water is unavailable.

Thus, **F1 should be irrigated automatically, whereas F3 should generate an alert or wait for irrigation availability.**

4. Analyze whether predicate logic alone is sufficient for agricultural decision support.

Answer:

Predicate logic is useful for representing **clear and deterministic facts**, such as whether soil is dry or irrigation is available. However, it is not sufficient when sensor readings are incomplete, noisy, or uncertain.

For example, a moisture sensor may provide an uncertain reading due to sensor errors. Predicate logic generally represents facts as true or false and cannot naturally express degrees of uncertainty.

Therefore, agricultural systems can combine predicate logic with **fuzzy logic, probability, Bayesian reasoning, and machine learning** to make better decisions under uncertain conditions.

3. Word Sense Disambiguation in a Travel Recommendation System

1. Determine the intended sense of the ambiguous terms using the available context.

Answer:

Word Sense Disambiguation (WSD) determines the correct meaning of an ambiguous word based on its **context and user behavior**.

Query	Intended Sense	Supporting Evidence
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages.
Safari booking	Wildlife tour	User selected wildlife resort packages.
Java vacation	Java island	User viewed hotels in Bali and Java.
Cruise package	Ship journey	User viewed Mediterranean ship packages.
Safari download Windows	for Safari browser	web “Download”, “Windows”, and browser page indicate software.

Thus, the correct sense is determined by combining the query with the user's interaction history.

2. Identify the lexical and contextual cues that distinguish the possible senses.

Answer:

Different words surrounding an ambiguous term provide important clues.

- **Amazon:** “tour”, “rainforest”, and “trekking” indicate the rainforest, while “shopping”, “Prime”, and “orders” would indicate the company.
- **Safari:** “booking”, “wildlife”, and “resort” indicate a wildlife trip, while “download”, “Windows”, and “browser” indicate software.
- **Java:** “vacation”, “hotel”, “Bali”, and “island” indicate the geographical location, while “programming”, “code”, and “JDK” indicate the programming language.
- **Cruise:** “Mediterranean”, “ship”, and “journey” indicate a travel cruise, while software-development terms may indicate another meaning.

Therefore, both **lexical context and user behavior** are important for WSD.

3. Analyze why the same word may require different interpretations for different users.

Answer:

The meaning of an ambiguous word depends on the **context, user's interests, previous searches, and current activity**.

For example, the word “**Safari**” can refer to a wildlife trip or a web browser. A user searching “**Safari booking**” is likely interested in wildlife travel, whereas a user searching “**Safari download for Windows**” is referring to the browser.

Therefore, WSD systems should not depend only on the dictionary meaning of a word. They must consider **personalized context and user behavior**.

4. Design an industrial-scale WSD strategy.

Answer:

An industrial-scale WSD system can use multiple sources of information:

1. **Query Context:** Analyze words surrounding the ambiguous term.
2. **User History:** Examine previous searches, clicks, bookings, and viewed products.
3. **Embeddings:** Use contextual word embeddings or transformer models to understand the meaning of the word in context.
4. **Click Behaviour:** Use clicked results and browsing behavior as evidence of the intended sense.
5. **Sense Ranking:** Generate possible meanings and assign a probability or score to each one.
6. **Continuous Learning:** Update the model using new user interactions and feedback.

For example, if the query contains “**Safari + wildlife + booking**”, the wildlife-travel sense receives a high score. If it contains “**Safari + download + Windows**”, the browser sense receives a higher score.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. Analyze how syntactic structure influences semantic-role assignment.

Answer:

Syntactic structure helps determine the relationships between the **subject, verb, object, and other sentence components**. However, the grammatical subject is not always the semantic Agent.

In the sentence:

“The supplier delivered the equipment to the buyer.”

- Supplier = **Agent**
- Equipment = **Object**
- Buyer = **Recipient**

In the passive sentence:

“The contract was terminated by the company.”

- Contract = **Patient/Object**
- Company = **Agent**

Thus, semantic analysis must consider **active/passive voice, dependency relationships, and prepositional phrases** rather than simply assigning the Agent role to the grammatical subject.

2. Identify the incorrect semantic-role assignments.

Answer:

There are two incorrect interpretations.

Sentence 2:

“The buyer received the equipment from the supplier.”

The system gives:

Buyer = Agent, Equipment = Object, Supplier = Recipient

This is incorrect.

The correct interpretation is:

- **Buyer = Recipient**
- **Equipment = Object**
- **Supplier = Source/Agent**

The phrase “**from the supplier**” indicates the source of the equipment.

Sentence 4:

“The contract was terminated by the company.”

The system gives:

Contract = Agent, Company = Object

This is incorrect because the sentence is in **passive voice**.

Correct interpretation:

- **Company = Agent**
- **Contract = Patient/Object**

The company performs the termination action, while the contract is affected by it.

3. Explain how confusing grammatical subject with semantic Agent affects legal information extraction.

Answer:

A grammatical subject does not necessarily perform the action. This is especially important in **passive constructions**.

For example:

“The contract was terminated by the company.”

Here, **contract** is the grammatical subject but it is not the Agent. The **company** is the actual Agent.

If a legal NLP system treats every subject as an Agent, it may extract incorrect information about **who performed an action, who received an obligation, or who was affected by an event**. This can lead to incorrect contract analysis and unreliable legal information extraction.

4. Recommend a syntax-driven semantic analysis architecture.

Answer:

A reliable legal NLP system should use the following architecture:

1. **Text Preprocessing:**

Perform tokenization, sentence segmentation, POS tagging, and named-entity recognition to identify legal entities such as companies, people, contracts, and dates.

2. **Syntactic Parsing:**

Use dependency or constituency parsing to identify subjects, verbs, objects, and prepositional phrases.

3. **Voice Detection:**

Identify whether a sentence is active or passive and detect **“by” phrases** in passive constructions.

4. **Semantic Role Labeling:**

Assign roles such as **Agent, Patient, Object, Recipient, Source, Instrument, and Location**.

5. **Dependency and Context Analysis:**

Handle prepositional phrases, embedded clauses, and long-distance dependencies using dependency parsing and contextual language models.

6. **Legal Validation:**

Apply domain-specific rules to verify whether the extracted semantic relationships are logically consistent.

Conclusion

A combination of **syntactic parsing + semantic role labeling + voice detection + contextual language models + legal-domain rules** provides a reliable approach for extracting accurate semantic information from complex legal documents.